

WHAT IS THE SFMA?



The Selective Functional Movement Assessment (SFMA) is a comprehensive assessment used to classify movement patterns, identify regions for further local biomechanical examination, and ultimately, direct manual therapy and therapeutic exercise interventions.

The SFMA is a system that helps rehabilitation professionals incorporate a movement baseline into their musculoskeletal examination. It does not replace impairment measurements or other specialized testing. It simply creates a more pattern-specific perspective of how each individual patient moves in relationship to their anatomical information and their medical diagnosis.

Anatomical information means that each patient has specific structural integrity that must be considered. Each patient also has a medical diagnosis that involves both indications and contraindications. However, anatomical information and medical diagnosis are not enough information to prescribe interventions.



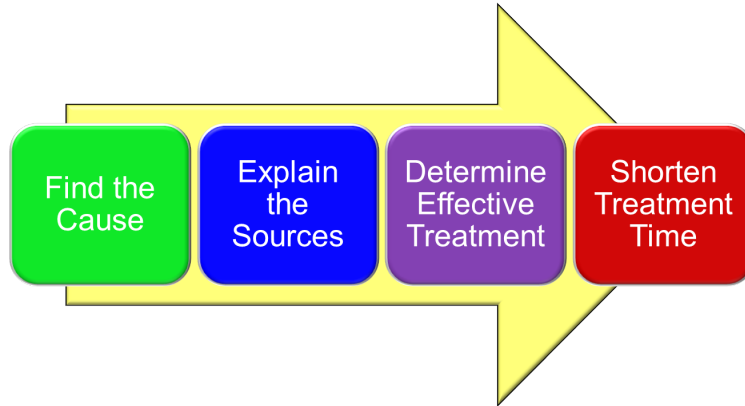
The SFMA allows the clinician to identify both movement dysfunction and pain specific to the individual they are assessing. It is easy to see how three back pain patients can have three different movement profiles - each demonstrating different levels of competency with mobility and stability. Likewise, three different total knee patients can demonstrate three different levels of competency with single-leg stance, forward bending, backward bending, and rotation. This identifies the importance of looking beyond the medical diagnosis and/or patient's structure when formulating the most efficient and effective plan of care. As rehabilitation professionals, we no longer "guess" at the cause of the patient's pain - the SFMA gives us a complete differential diagnosis. Often the cause of the pain is not the source of the pain.

The SFMA also creates an opportunity for reappraisal. This means, when we treat a specific area of the body and work on a specific impairment, we can quickly see if that intervention changed local movement competency and also had an effect on global, or whole pattern, movement competency. The SFMA acts as a metric for movement and a metric for change - did our intervention actually change movement?

DIAGNOSTICS

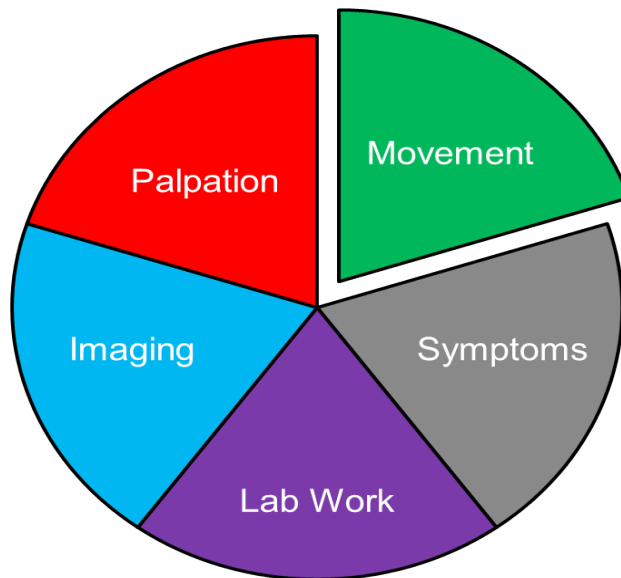


What is the Purpose of Diagnostics:



What are Your Options for Diagnostics:

We now know the traditional isolated approach to either evaluation or treatment will not always restore function. Therefore, we need to use better diagnostics - so what are our options?



The SFMA uses movement as its form of diagnosis.

SFMA KEY CONCEPTS



Selective Functional Movement Assessment Key Concepts

CONCEPT 1: REGIONAL INTERDEPENDENCE

“This refers to the concept that seemingly unrelated impairments in a remote anatomical region may contribute to or be associated with the patient’s primary complaint.”

This is the hallmark of the SFMA as we consider the concept of regional interdependence that tells us the motor system must be considered in its entirety. The traditional medical model would consider each individual component in and of itself in isolation. Our medically oriented training teaches us how to do a standardized orthopedic examination. This exam is done primarily with the patient on the table looking at the area of the primary complaint and, at the very most considering a joint above or a joint below and comparing to the opposite side. As we embrace the concept of regional interdependence, we must consider the entire movement system as a whole. We start with our top tier movements, considering body movement regardless of the patient’s region of complaint.

Vladimir Janda, the late physician and physical therapist, was thinking in these terms well ahead of his time and, in fact, quotes and research related to this date back to the mid 1960’s:

“The motor system functions as an entity. It is principally a wrong approach to try to understand impairments of different parts of the motor system separately without understanding the function of the motor system as a whole.”

Janda goes on to discuss how we would typically assess muscle function and joint dysfunction.

“Only gross changes in muscle function in the sense of weakness and exceptional shortening and tightness are emphasized. There are almost no data about the analysis of movement patterns”.

Janda was one of the first rehabilitationists to note the importance of assessing movement patterns, as a whole, in adults without neurological disorders. His original work on patients with chronic LBP suggested that we put things together to assess the neuromuscular system in its entirety.



SFMA KEY CONCEPTS



The way that Gray Cook first described a similar concept in 2001:

“The observation of whole movements may redirect and broaden the clinical focus by revealing limitations unrelated to the medical diagnosis but pertinent to restoration of normal function”.

For the rehabilitation professional, we are suggesting that the medical diagnosis will not in and of itself guide the rehabilitative treatment. We must consider whole patterns of movement and be sure that they are assessed for every patient.

The medical diagnosis is, of course, important and we do want to confirm the medical diagnosis. But in reality, the medical diagnosis will provide us contraindications to treatment rather than a treatment plan. For example, the medical diagnosis will help us with healing times, movement constraints and specific restrictions to the rehabilitation process, but will not help us to specifically prescribe manual therapy and exercise.

Another way to consider this concept is the idea of “Source vs. Cause”. The medical diagnosis and traditional orthopedic special testing will uncover the “source”. But the “cause” is often mobility or motor control deficits in another region or regions and is hard to predict based on only the medical exam. The medical diagnosis will provide the “source” and the SFMA diagnosis will provide the “cause”.

The SFMA, in part, is based on the concept of regional interdependence in that we must assess and treat dysfunction away from the patient’s primary location of pain. The SFMA will guide the clinician to the most dysfunctional region that is adversely affecting the movement pattern. There are a number of studies supporting this. In fact, as Rob Wainner stated in his 2007 commentary on regional interdependence, when asked the question, “What evidence is there to suggest that an orthopedic model for regional interdependence is required or is needed?” his answer, “Surprisingly, plenty”.

There are numerous studies that have linked dysfunction at one area of the body with pain and dysfunction in another area of the body. The most common of these examples include hip impairments that are related to low back pain and thoracic spine impairments related to neck pain and shoulder pain. In a follow-up paper by Wainner and colleagues in 2013, several more studies are mentioned demonstrating the relationship between treating regions away from the main area of pain and associated functional improvements. In short, regional interdependence provides the theoretical support to assess and treat a non-affected body part. Rehabilitation professionals are the most highly trained healthcare professionals to function within this model.

SFMA KEY CONCEPTS



Regional Interdependence: Mobility/Stability Alternating Pattern

One way to look at regional interdependence is the relationship between mobile and stable segments. If you consider any joint that has its primary movement in one plane of motion a stable joint, and those that don't have one primary plane of movement a mobile joint, you can make the following observation (first noted by Mike Boyle and Gray Cook):

“The body works in an alternating pattern of stable segments connected by mobile joints. If this pattern is altered – dysfunction and compensation will occur.”

Normal Pattern

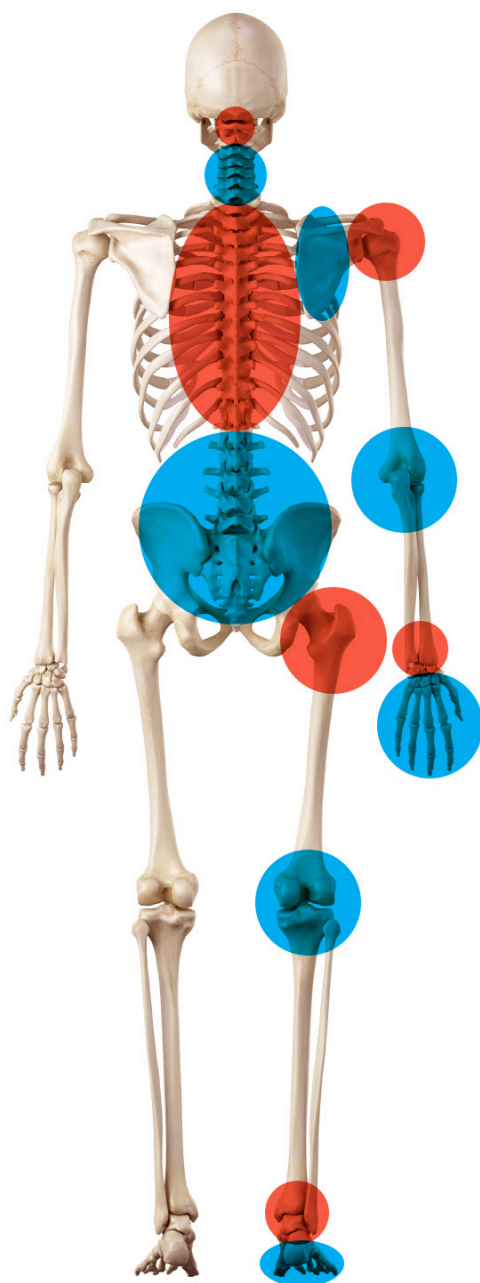
Foot	Stable	
Ankle	Mobile	
Knee	Stable	
Hip	Mobile	
Pelvis/Sacrum/Lumbar Spine	Stable	(See Diagram on next page)
Thoracic Spine	Mobile	
Scapulo-Thoracic	Stable	
Gleno-humeral / Shoulder	Mobile	
Elbow	Stable	
Wrist	Mobile	
Cervical Spine	Stable	

You can see how this observation of an alternating pattern of mobility and stability can help describe how injuries occur. If you take the lower back as an example, you will often find that the hip joints and thoracic spine are limited in mobility. Therefore, the lumbar spine will sacrifice stability to obtain more motion. This abnormal motion in the lumbar spine can be one of the primary reasons for disc and facet injuries in the lower back.

SFMA KEY CONCEPTS



Regional Interdependence: Mobility/Stability Alternating Pattern



STABILITY

MOBILITY

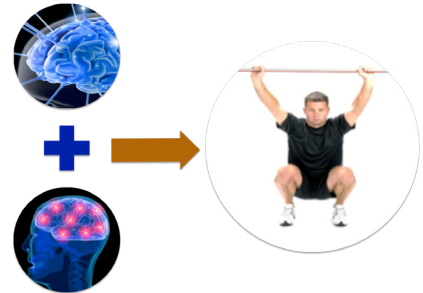
SFMA KEY CONCEPTS



CONCEPT 2: ALTERED MOTOR CONTROL

We should first address the question.....What is Motor Control? Our simple, yet comprehensive definition for motor control is:

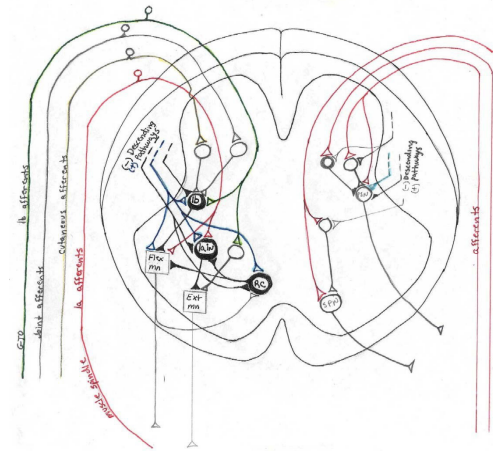
“Necessary input, sufficiently processed, with an acceptable output”



Necessary Input

Involves on-going sensory information gained from the periphery passed to the CNS through 4 primary pathways:

- » Golgi Tendon Organs
- » Joint Afferents
- » Cutaneous Afferents
- » Muscle Spindles



Sufficiently Processed

Afferent information is processed appropriately and blended with information from descending pathways to create and refine movement. Appropriately implies the information is processed free of noise from the movement system having to deal with other stressors such as breathing dysfunction, pain, emotional stresses or fear.

Acceptable Output

Because it is challenging to accurately assess or measure the *input* and the *processing*, we start by assessing the output. In the case of the purpose of the SFMA we start with an “acceptable” movement pattern. So, meeting the criteria of the top tier movements as described or the criteria of a given breakout test would satisfy that an acceptable output has been observed. As we move into other, more demanding components of the Functional Movement System, the range of acceptable outputs becomes more challenging such as the 7 movements of the Functional Movement Screen, or the body relative expectations of single limb dynamic balance on the Y-Balance Test. Each component has minimum movement standards that must be met to consider adequate motor control is present to allow the opportunity for the patient/client to learn and mature from a movement perspective in more challenging environments.

SFMA KEY CONCEPTS



“Mobility before Motor Control”

We emphasize that intervention should begin with the treatment of mobility deficits prior to addressing deficits in motor control. We emphasize this because it honors our definition of motor control. If a given joint/soft tissue unit lacks fundamental mobility, the potential for ideal motor control is by definition limited. That is, lack of mobility will reduce afferent input, creating a situation where ideal motor control cannot be achieved. With mobility deficits managed first, the opportunity for ideal motor control exists, but is not guaranteed.

The term motor control is important because it the operating system that “grows” our movement abilities. Our CNS is literally wired to learn and adopt the movements that match our motivations from the most basic to the most complex and specialized. We express our movements in the form of patterns and they are managed by the CNS.

What happens to movement following injury?

This, of course is a multifaceted question, but we have learned that the response to movement following injury is more complex than previously thought. One approach is to look at injury risk factors to, in a sense, work backwards to help us answer the question. By considering risk factors for injury, we gain a better understanding of what is happening in the motor control system after injury.

It is clear from the peer reviewed literature that previous injury is by far the most robust factor related to future injury. With this fact in mind, it should make us feel somewhat uncomfortable as rehabilitationists that those with a previous injury, even after completing rehabilitation, are at the greatest risk of subsequent injury.

Additional findings related to future injury include neuromuscular impairments that have identified a variety of types of asymmetries, either right-left movement differences or muscle imbalances, as well as a broader area that we consider under the umbrella of balance and motor control deficits. There are many prospective studies that have found that one or more motor control deficits are related to injury. We will discuss just a few studies (specifics) so that you can get a better sense of what this is like. First, we present a summary of studies of patients post ankle sprain or with chronic ankle instability, who were pain free at the time they were tested. In these subjects, a variety of impairment measures have been found including decreased hip and knee joint angles and decreased overall reach distance with the Y-Balance Test. This indicates that those who have had an ankle sprain perform poorly on a simple dynamic balance test even though they are pain free and back to sport/activity.

In a large prospective study on army soldiers, several risk factors for injury were identified. Of particular interest was that ankle dorsi-flexion (closed chain measure in half kneeling) asymmetry of 5 degrees or greater was a proven risk factor. This reduced/asymmetrical motion was a predictor for any time-loss injury going forward, not specific to just ankle injuries. This suggests that reduced/asymmetrical ankle mobility may alter motor control creating an increased risk of injury.

SFMA KEY CONCEPTS



Next are studies from the biomechanical literature that look at EMG data including onsets of activation of a variety of muscles. These data indicate that the onset of activation in the gluteus maximus in an ankle sprain patient is delayed or comes on slower when compared to a control group and significant latencies have been demonstrated in the gluteus medius. Again, remember these subjects were pain free at the time they are tested. This brings us back to the whole concept of regional interdependence that an injury at the ankle actually results in altered gluteus function.

The most profound data, is from a study looking at motor control during simply transitioning from double limb stance to single limb stance. During even this very low-level task, significant delays in the onset of ankle, hip and hamstring musculature was noted when compared to controls. It appears that subjects adapt a substantially altered strategy of movement, which is retained by the motor control system, even after the subject has returned to full activity. This adaptation serves as a profound example of motor control changes and altered motor control that occurs following injury. It is possible that standard rehabilitation programs simply do not get to the level where we can address these motor control changes.

When looking further into movement patterns, one study showed clearly that movement patterns are altered following substantial injury and that they may not resolve with standard rehabilitation. This study looked at total knee arthroplasty patients. After their rehabilitation and one year after their total knee, motion analysis data showed that they still relied on altered hip and ankle strategies when performing a simple sit-to-stand task. This tells us that a quad avoidance strategy persists one year after total knee arthroplasty, even though quad strength has resolved. "This pattern may be a learned movement pattern that may not resolve without retraining".

We know that with a musculoskeletal injury that there are changes in motor control. These are broad statements, but they are the driving concepts behind why it is so important that we look at full movement when the goal is complete restoration of function. Injury alters motor control. This, in part, may help explain why previous injury is the most profound risk factor for future injury.

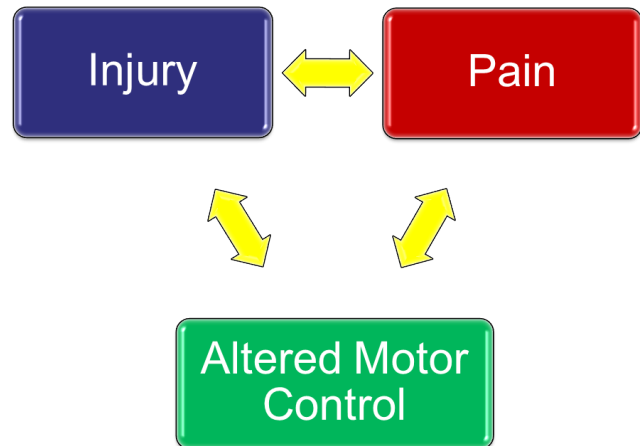
SFMA KEY CONCEPTS



Pain and Motor Control

The next important portion to consider related to motor control changes is pain. Pain alters motor control in a somewhat unpredictable manner. Initial assumptions of pain and motor control stemmed from basic inhibition – that is with joint pain, there was inhibition of the muscles in the area -- a simple reflex inhibition. A common example is when the VMO is shut down due to knee effusion and/or pain.

Further research has shown that pain does not always decrease muscle activation. In fact, when pain exists, you can occasionally get an increase in muscle activation. The Pain Adaptation Model proposed by Lund in 1991 predicted that pain would alter motor control depending on the given role of a muscle as an agonist or antagonist. This model has become useful in some situations but has been disproven in others and as more research has been performed to better understand how pain alters motor control.



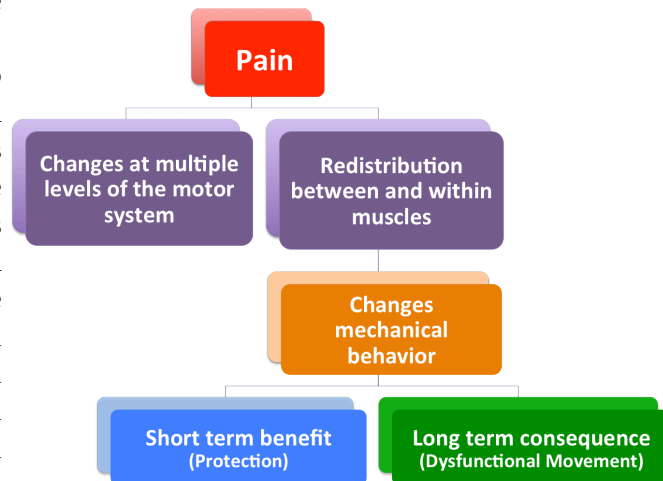
Current thinking suggests that pain alters motor control in a somewhat unpredictable manner and that it is most likely related to the particular task at hand. Depending on the given task, when pain is present, the system may increase or decrease muscle activation. This was similar to what we found in a study that we did looking at lumbar multifidus function during a simple arm lifting movement and a simple forward and backward weight shifting movement. These were performed under three conditions – the normal condition with no pain, after pain had been induced with hypertonic saline, and then following recovery. What we found was that during the arm lifting movement, the central nervous system increased activation to the lumbar multifidus, and during the weight shifting movement, muscle activation was decreased. The system is very adaptable and can alter muscle activation very quickly – immediately in this case, likely driven by the task. We make this point to recognize that there are many ways to influence the movement system. We will refer to a cross section of the spinal cord where we see afferents pathways information coming in from muscle spindles, Golgi tendon organs, joint afferents throughout the joint and cutaneous afferents from the skin. Each pathway comes in and will synapse with interneurons and synapse with fibers from the descending pathways from multiple levels of the central nervous system. This descending influence can come all the way up from the motor planning levels of the cortex. Eventually all of that information comes together to send a signal to the muscle to either increase or decrease its activity.

This is a very complex system, and no one quite understands completely how it works, but with four competing theories of motor control that have come to the forefront today, what we want to understand from a rehabilitation standpoint is many things we do influence movement. It is important to recognize we have a lot of ability to immediately change motor control.

SFMA KEY CONCEPTS



There are lots of new ways we are looking at quickly changing motor control. We do this through our prioritization exam. With pain and dysfunction, it is important to understand what happens to motor control, not only in general theoretical terms but specifically what happens at the core. Core training and core muscle function are important and very popular. What is important as rehabilitationists is we understand a simple strengthening model; a “more is better” mentality is not always the best way to approach core training. A series of research studies have shown when someone has a history of low back pain their core function is altered. This series of studies by Choliwicki has taught us athletes with a history of low back pain, pain free at the time they were tested, demonstrate altered core muscle control. These studies looked at global muscles - the large two joint and multi-joint muscles around the core – and showed when a load is suddenly removed from the spine at an unpredictable time, those who have a history of back pain tend to leave on their core muscles longer and leave them on at higher EMG levels when the load is removed. Instead of fine control of the core, those who have had a low back pain episode tend to use a gross co-contraction type strategy which is less specific, less controlling, with higher muscle activation in the superficial core musculature.



These data have helped us understand, in one small window of time during this sudden unloading, how subconscious core muscle function occurs and how core muscle function is altered after someone has had a back pain episode. This data is very important to understand before prescribing traditional core training.

Types of Pain

It is important to recognize the contemporary definitions of pain. Modern pain science suggests there are 3 types of pain that are most commonly encountered in current clinical practice: nociceptive pain, peripheral neuropathic pain and central pain.

It has been suggested that the majority of pain managed related to musculoskeletal system is nociceptive in nature. The manner in which we utilize pain as a category related to movement is consistent with this primary definition. For the purposes of the SFMA, following the categorical approach to movement patterns has been designed around the nociceptive definition of pain. For patients with peripheral neuropathic pain, SFMA principles can be applied. For patients with central pain, this can be considered a processing deficit. Despite near normal or normal input, the motor output remains problematic. There are known clinical approaches described elsewhere to address those with central pain from a behavioral perspective. Here are the definitions and key clinical characteristics of the 3 pain types as described by Smart et al 2010.

SFMA KEY CONCEPTS



Nociceptive pain mechanisms refer specifically to those pathophysiological processes associated with activation of the peripheral receptive terminals of primary afferent neurons in response to noxious chemical (inflammatory), mechanical or thermal stimuli. Chemically mediated nociception from the inflammatory response associated with tissue injury or from tissue ischemia in response to tissue loading or compression are likely mechanisms underlying many clinical presentations of musculoskeletal pain.

Characteristics of Nociceptive Pain:

- Clear, proportionate mechanical/anatomical nature to aggravating and easing factors.
- Pain associated with and in proportion to trauma or a pathological process or movement/postural dysfunction.
- Pain localized to the area of injury.
- Resolving in accordance with expected tissue healing/pathology recovery times.
- Responsive to simple analgesia/NSAID's.
- Pain in association with other symptoms of inflammation (i.e. swelling, redness, heat) (Inflammatory nociceptive).

Peripheral neuropathic pain has been defined as “Pain initiated or caused by a primary lesion or dysfunction in the peripheral nervous system” and describes a mechanistic category of pain involving a number of pathophysiological mechanisms associated with altered nerve functioning and responsiveness, such as changes in electrical hyper-excitability and abnormal impulse generation, and mechanical, thermal and chemical sensitivity.

Characteristics of Peripheral Neuropathic Pain:

- Pain variously described as burning, shooting, sharp, aching or electric-shock-like.
- History of nerve injury, pathology or mechanical compromise. Pain in association with other neurological symptoms (e.g. pins and needles, numbness, weakness).
- Pain referred in a dermatomal or cutaneous distribution. Less responsive to simple analgesia/NSAID's and/or more responsive to anti-epileptic (e.g. Neurontin, Lyrica)/anti-depression (e.g. Amitriptyline) medication.
- Pain of high severity and irritability (i.e. easily provoked, taking longer to settle).
- Mechanical pattern to aggravating and easing factors involving activities/postures associated with movement, loading or compression of neural tissue.
- Pain in association with other dysesthesias (e.g. crawling, electrical, heaviness).

SFMA KEY CONCEPTS



Central pain has been defined as “Pain initiated or caused by a primary lesion or dysfunction in the central nervous system (CNS).” The term “central” pain was used in this study to refer to musculoskeletal pain states characterized primarily by dysfunction within the CNS i.e. central sensitization/hyper-excitability rather than by any distinct lesion within the CNS as might occur with CNS disorders such as multiple sclerosis.

Characteristics associated with Central Pain:

- Disproportionate, inconsistent, non-mechanical/non-anatomical
- Pattern of pain provocation in response to movement/mechanical testing.
- Positive findings of hyperalgesia (primary, secondary) and/or allodynia and/or hyperpathia within the distribution of pain. Diffuse/non-anatomic areas of pain/tenderness on palpation. Positive identification of various psychosocial factors (e.g. catastrophisation, fear-avoidance behaviour, distress).

CONCEPT 3: NEURODEVELOPMENTAL SEQUENCE

Because of the changes in motor control that occur with injury and pain, we have adopted a model that utilizes the basics of the neurodevelopmental progression to help guide our examination and intervention techniques.

The neurodevelopmental progression has taught us that movement was developed in patterns and not in individual muscles. This goes back to Janda’s point of assessing patterns of movement. By considering the neurodevelopmental progression as our basis for movement, we are able to capture nature’s method of developing movement, which we refer to as “authentic movement”. We revisit the basic neurodevelopmental postures during assessment and correction of movement.

The normal sequence of learning movement is the following:

- Breathing
- Grasping/Gripping
- Head and Eye Movement
- Limb Movement
- Rolling
- Crawling
- Kneeling
- Transitional Movements
- Standing

Our examination and exercise progressions build on changes in neurodevelopmental postures. We may start with something as simple as diaphragmatic breathing in the supine or prone position and then progress from there to prone-on-elbows, then to quadruped, quadruped to kneeling, kneeling to half kneeling, half kneeling to standing and ultimately getting to a single leg stance activity, for example the single leg dead lift. We are careful to assess breathing patterns across the neurodevelopmental progressions as we move to the higher level of exercises. Understanding that part of the compensatory changes that occur with pain and motor control involve core motor control changes as discussed. One of the primary roles of the deep core is breathing. Often, we will find patients in a slightly distressed state that is manifested in a simple apical or

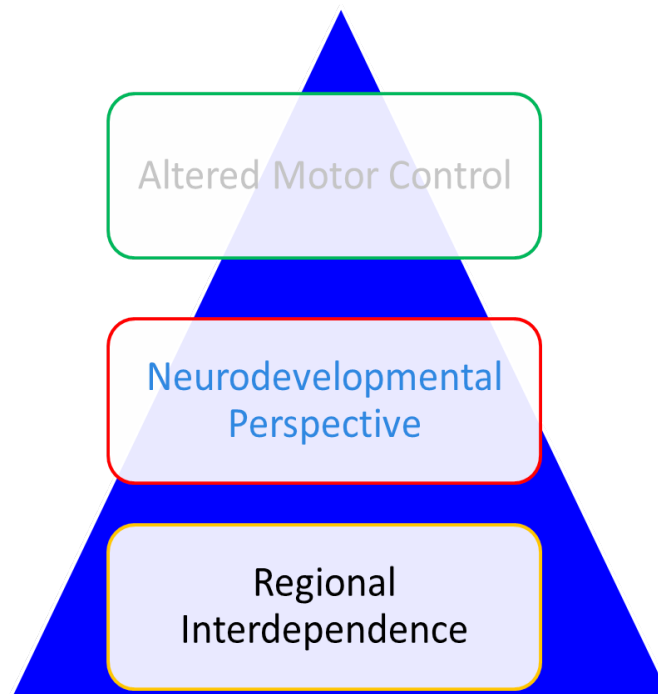
SFMA KEY CONCEPTS



upper chest breathing pattern. We are quick to correct this through volitional training with hands-on assistance and again assess breathing throughout every neurodevelopmental posture. We simply utilize the concepts of authentic movement to guide how adults should move.

“Poor movement can exist anywhere in the body. Poor movement patterns exist only in the brain.”

This quote helps us understand the importance of motor learning, motor re-learning and movement pattern training as we move forward with our model. We recognize that the body will always sacrifice quality over quantity of movement, and that there is a certain amount of compensatory movement that is tolerable. However, once we cross a particular threshold, that compensatory movement becomes dysfunctional movement. We feel that dysfunctional movement – whether a result of pain or injury or causing pain or injury – (you can look at it either way) is one of the missing links related to moving rehabilitation and injury prevention science forward.



SCREEN vs. TEST vs. ASSESSMENT



What is a Screen?

1. A system for selecting suitable people.
2. To protect somebody from something unpleasant or dangerous.

Basically, this means - To check risk!

What is a Test?

1. A series of questions, problems, or practical tasks to gauge somebody's knowledge, or experience ability.
2. Measurement with no interpretation needed.

Basically, this means - To measure ability!

What is an Assessment?

1. To examine something in order to judge or evaluate it.
2. To calculate a value based on various factors.

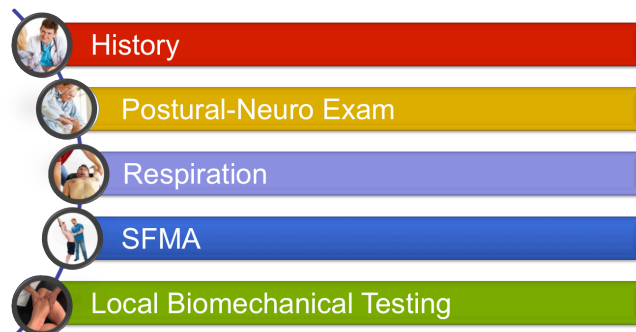
Basically, this means - To judge inability!

Our Working Definition:

- Screen - Risk
- Test - Ability / Function
- Assess - Inability / Dysfunction

The Functional Orthopedic Examination:

1. History & Observation
2. Postural-Neuro Exam (UQS or LQS)
3. Breathing
4. SFMA
5. Clinical Testing
 - Palpation
 - ROM: active & passive
 - Muscle assessment
 - Joint mobility/stability
 - Special Testing



Start with the SFMA prior to the specific clinical assessment simply to direct the evaluation in an appropriate and constructive manner (Rate & Rank). Refine, qualify, and quantify the functional parameters of the problem at hand before getting specific.

FUNCTIONAL MOVEMENT SYSTEMS



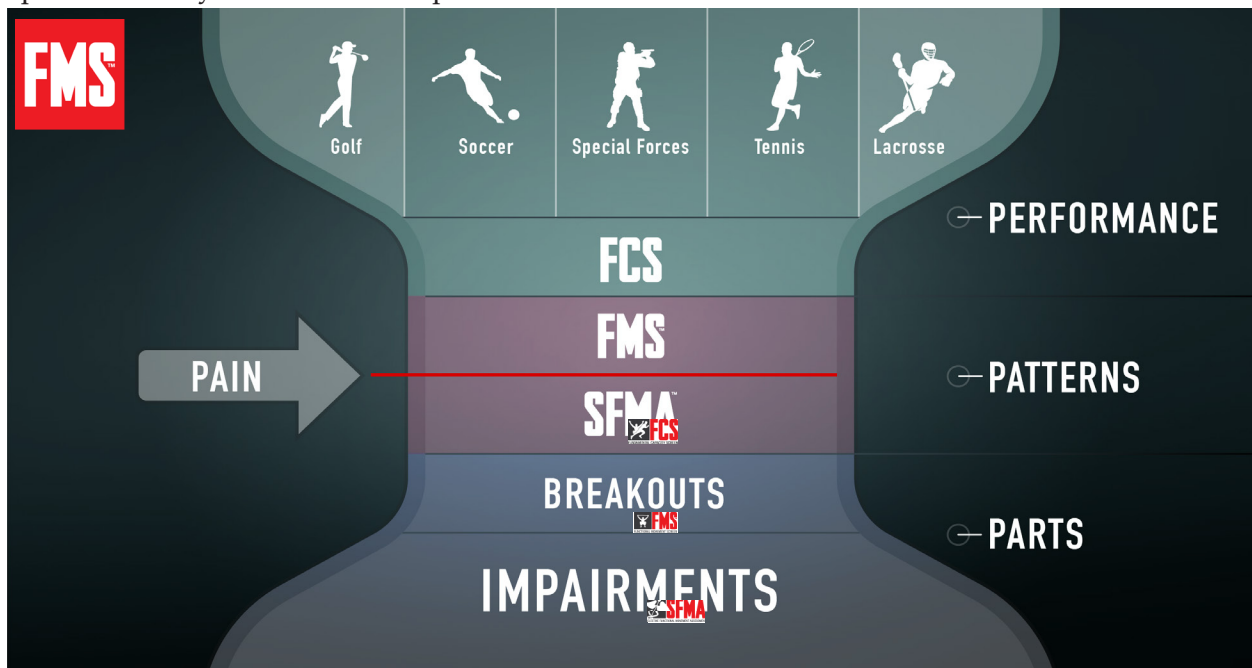
How to Start the Movement Evaluation for Performance

Many times, you will be asked to evaluate an individual to see if you can improve overall performance. They have no complaint of pain, they just want to either prevent problems from occurring or improve their overall performance. The evaluation of this client should begin with a quick movement screen.

We recommend beginning with the Functional Movement Screen (FMS). This screen was designed to evaluate patterns of movement and identify or highlight altered patterns in pain free individuals. If the FMS produces pain, then the screen needs to be progressed to a thorough assessment - that is the SFMA. The SFMA will help identify the cause of the pain and direct the treatment of the individual. Once the pain has been resolved, the FMS should be used to re-evaluate the patterns of the individual. If the FMS is dysfunctional but not painful, you should focus on restoring the altered patterns that were discovered.

If the FMS was normal and no altered patterns were detected, then the next level of evaluation should include capacity testing. Before we learn and acquire occupational, recreational or sport specific skills, we express energy through elemental capacities that can, and should, be measured. Capacity testing allows us to identify “leaks” in an individual’s ability to express energy.

The next step in the progression is to assess skill and/or specific characteristics needed for an activity. This will allow the evaluator to see how the athlete or patient performs in their specific sport or activity. Deficiencies in performance should be addressed at that time.



SFMA



Much like a compass points a traveler toward a destination, this qualitative assessment (SFMA) gives direction to the problem-solving process used in a musculoskeletal evaluation. Once the traveler is headed in the right direction, then the quantitative data - like time, speed and distance - has relevance. If, however, the traveler was headed in the wrong direction, then this data is of little use to reaching the desired destination. Often the novice clinician will collect a large amount of quantifiable data without ever identifying the nature of the fundamental problem, much like the traveler who might be making excellent time, but in the wrong direction.

The SFMA should be thought of as your baseline pain and injury assessment. It will allow you to systematically identify the patterns of movement that are contributing to the patient's pain and help guide you in the proper direction for local biomechanical examination and eventually, treatment.

The SFMA is a functional movement ranking system. The system directs the clinician to rank pain and dysfunction for basic movement patterns. Patterns are broken down to terminal points or until pain or dysfunction is no longer observed. The system helps the user to rank movement behavior prior to clinical measurements of strength, ROM, joint stability, biomechanical tests, etc.

The Scoring System for the SFMA:

We will use the following to score movements within the SFMA:

- » **Functional** - this term will describe movement that is unrestricted or unlimited, meeting standard ROM's and quality of movement.
- » **Dysfunctional** - this term will describe movement that is limited or restricted in some fashion - demonstrating a lack of mobility, stability or symmetry.
- » **Painful** - this term will describe when the SFMA reproduces symptoms, increases symptoms, or brings about secondary symptoms. Additional notes may be used to describe the nature and severity of the pain experienced.
- » **Non-painful** - this term will describe movement that does not reproduce or produce pain.

Each movement within the SFMA must be scored with one of four possible answers:

1. Functional and Non-Painful (FN)
2. Functional and Painful (FP)
3. Dysfunctional and Painful (DP)
4. Dysfunctional and Non-Painful (DN)

THE TRAFFIC LIGHT SYSTEM



The Traffic Light System

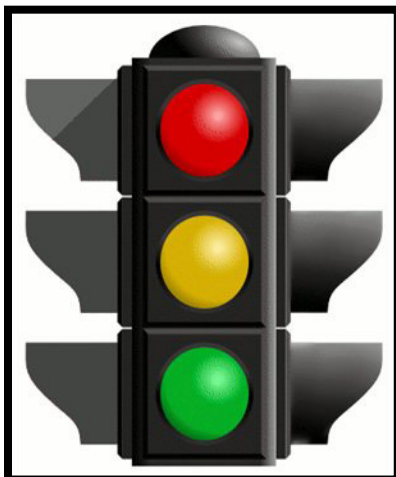
We have devised a color system to help guide you through the SFMA - indicating what avenues to continue exploring and which do not need to be further assessed. It uses the same colors that are found on a traffic light (red, yellow and green).

The Top Tier:

- **Red Light:** This means STOP - you do not have to continue with a breakout of that pattern as they demonstrated function. These patterns will be functional and non-painful (FN).
- **Yellow Light:** This means proceed with caution - you must breakout these patterns, but there is pain (FP or DP) involved so be careful. Use these breakout findings as indicators if your treatments are working (re-test them frequently).
- **Green Light:** This means go - you need to break these patterns out as demonstrated some form of dysfunction with the pattern. These patterns will be dysfunctional and non-painful (DN). These are the key to the SFMA in identifying what is contributing to the patient's pain and will lead you to the proper area for further local examination and ultimately treatment.

The Breakouts:

- **Red Light:** This identifies a normal finding and can direct you to look at another breakout.
- **Yellow Light:** This means you have identified a painful local region. It is appropriate to do a local biomechanical examination to diagnosis the cause of the painful dysfunction and then treat accordingly.
- **Green Light:** This means go - you have your answer on what is causing the dysfunction. It is appropriate to do a local biomechanical examination to diagnosis the dysfunction and then treat accordingly.



The Selective Functional Movement Assessment (SFMA) is a comprehensive assessment used to classify movement patterns and direct manual therapy and therapeutic exercise interventions.

RULES OF SCREENING



Some of the basic rules of screening:

1) No warm up - allows repeatability and evaluates raw, unaltered movement.



2) If it looks like a dog and smells like a dog - it's a dog!



3) Be picky - If it is “not bad” that means it is “not good”.



4) No shoes



5) Monkey see, monkey do - allows for faster evaluation times.



SFMA TOP TIER



SFMA SCORING		FN	FP	DP	DN
Cervical Flexion					
Cervical Extension					
Cervical Rotation	L				
	R				
Upper Extremity Pattern 1(MRE)	L				
	R				
Upper Extremity Pattern 2 (LRF)	L				
	R				
Multi-Segmental Flexion					
Multi-Segmental Extension					
Multi-Segmental Rotation	L				
	R				
Single-Leg Stance	L				
	R				
Arms Down Deep Squat					

SFMA TOP TIER CHECKLIST



Name:

Date:

Total Score:

Cervical Flexion

☐ Painful

- ☐ Can't touch sternum to chin
- ☐ Non-uniform spine curve
- ☐ Excessive effort and/or lack of motor control

Cervical Extension

☐ Painful

- ☐ Not within 10 degrees of parallel
- ☐ Non-uniform spine curve
- ☐ Excessive effort and/or lack of motor control

Cervical Rotation

☐ Painful Right

☐ Painful Left

- ☐ Right ☐ Left Chin/Nose not in line with mid-clavicle
- ☐ Right ☐ Left Excessive effort and/or appreciable asymmetry or lack of motor control

UE Pattern #1 – MRE

☐ Painful Right

☐ Painful Left

- ☐ Right ☐ Left Does not reach inferior angle of scapula
- ☐ Right ☐ Left Excessive effort and/or appreciable asymmetry or lack of motor control

UE Pattern #2 – LRF

☐ Painful Right

☐ Painful Left

- ☐ Right ☐ Left Does not reach spine of scapula
- ☐ Right ☐ Left Excessive effort and/or appreciable asymmetry or lack of motor control

Multi-Segmental Flexion

☐ Painful

- ☐ Cannot touch toes
- ☐ Sacral angle <70 degrees
- ☐ Non-uniform spine curve
- ☐ Lack of posterior weight shift
- ☐ Excessive effort and/or appreciable asymmetry or lack of motor control

Multi-Segmental Extension

☐ Painful

- ☐ Upper extremity does not achieve or maintain 170
- ☐ ASIS does not clear toes
- ☐ Spine of scapula does not clear heels
- ☐ Non-Uniform spine curve
- ☐ Excessive effort and/or lack motor control

Multi-Segmental Rotation

☐ Painful Right

☐ Painful Left

- ☐ Right ☐ Left Pelvis Rotation <50 degrees
- ☐ Right ☐ Left Torso rotation <50 degrees
- ☐ Right ☐ Left Excessive effort and/or lack of symmetry or motor control

Single-Leg Stance

☐ Painful Right

☐ Painful Left

- ☐ Right ☐ Left Eyes open <10 seconds
- ☐ Right ☐ Left Eyes closed < 10 seconds
- ☐ Right ☐ Left Loss of Height
- ☐ Right ☐ Left Excessive effort or lack of symmetry or motor control

Arms Down Deep Squat

☐ Painful

- ☐ Hips do not break parallel
- ☐ Cannot reach fists to ground within footprint
- ☐ Loss of sagittal plane alignment: Right ____ Left ____
- ☐ Excessive effort, weight shift, or motor control

SFMA BREAKOUTS



The SFMA Breakouts

The Selective Functional Movement Assessment breakouts systematically dissect each of the major pattern dysfunctions described in the previous chapter. The breakouts are performed to improve the efficiency of the SFMA decision tree. You should investigate all top-tier patterns scored as Dysfunctional and Non-Painful (DN), Dysfunctional and Painful (DP) or Functional and Painful (FP). It is essential to break out all DN's before testing the DP's and the FP's. You should test the painful patterns last since they can lead to further tissue inflammation and exacerbation of the symptoms. Breaking out the DP's and FP's can make further testing impossible or extremely uncomfortable for the patient.

The breakouts are mostly global, non-measured appraisals and are used to suggest the need for further clinical investigation. These tests prove a logical connection between functional movement and impairment measurements. The breakouts should never be considered a terminal point unless they provide negative information and/or no other risk factors are present.

The SFMA breakouts are organized in a systematic way using a consistent logic. This logic includes three possible steps:

1. **Remove body parts if possible** - this logic step allows us to begin to isolate the cause of the dysfunction. This can include unilateral versus bilateral movement and/or removing sections of the body from the movement being tested. If, by removing body parts, the movement becomes functional, we can deduce the removed body part is the main cause of the movement pattern dysfunction.
2. **Change the stability requirements** - this logic step includes changing the postural demands for the task being performed, such as moving from a loaded position to an unloaded position. If when stability requirements are changed, the movement becomes functional, it cannot be called a mobility dysfunction as they demonstrated full range of motion - therefore, it would be called a stability/motor control dysfunction. If no change was seen, we must continue on with our logic steps.
3. **Active versus Passive movements** - this logic step allows us to differentiate between a mobility dysfunction and a stability/motor control dysfunction. If the movement is demonstrated passively, it cannot be a mobility dysfunction. If the movement is still restricted passively, your conclusion can be a mobility dysfunction for that movement. However, these mobility problems must be further examined with specific local testing.

This SFMA Logic will lead us to our differential diagnosis for each dysfunctional pattern. With all findings, the SFMA guides you to the proper region needing a further local biomechanical exam to confirm normal range or that you clinically measure the level of mobility impairment. The biomechanical exam should also include appraisals of structural and neuromuscular integrity as well as motor control. The biomechanical exam should indicate if impairment is present or absent and help complete the functional diagnostic process.

Local biomechanical testing is beyond the scope of the SFMA Level 1, but is used to specifically measure mobility with goniometric measurements, assess structural integrity with manual muscle testing and evaluate neuromuscular integrity and motor control. SFMA Level 2 includes local biomechanical testing.

SFMA TERMINOLOGY



The SFMA breakout testing applies the same categorizations as the top-tier assessment (FN, FP, DN, DP) with isolated focus in each pattern demonstrating pain or dysfunction. This focus helps the clinician identify gross limitations in mobility and stability and reach a differential diagnosis.

Unfortunately, the terms mobility and stability/motor control are not universally defined and can imply different things to clinicians of different backgrounds and training. Within the SFMA, the following definitions will be used:

Mobility Dysfunction (MD)

Mobility dysfunctions can be defined as a decrease or limitation in full range of motion. Limited mobility can be a result of either restrictions of the joint or restrictions within the tissue. The list of possible causes is quite long - the SFMA takes you to the area, your local biomechanical examination will identify the exact source of the problem.

The following are just some examples of mobility dysfunctions:

- Active/Passive Muscle Insufficiency
- Neural Tension
- Fascial Tension
- Muscle Shortening
- Hypertrophy
- Trigger Point Activity
- Scarring and Fibrosis
- Osteoarthritis
- Osteoarthrosis
- Uniarticular Muscle Spasm/Guarding
- Fusion
- Subluxation
- Adhesive Capsulitis

Stability/Motor Control Dysfunctions (SMCD)

An SMCD is present when the underlying mobility to complete the desired movement is present, but because of an input or processing problem the coordination of the movement is not demonstrated. Within the SFMA, we reach an SMCD finding when the patient demonstrates full mobility of the region, either actively when postural requirements are changed or with passive testing.

Traditionally, stability dysfunction is often addressed by simply attempting to concentrically strengthen the muscle groups identified as stabilizers. This approach neglects to consider that true stabilization is reflex driven and relies on proprioception and timing rather than isolated gross muscular strength. By using the term SMCD to distinguish stability/motor control problems, the clinician is forced to consider the central nervous system, peripheral nervous system, motor programs, movement organization, timing, coordination, proprioception, joint and postural alignment, structural instability, and muscular inhibition as well as the absolute strength of stabilizers. This new moniker also forces the clinician to consider situations like high threshold

SFMA TERMINOLOGY



strategy that use global muscles to accomplish tasks more suited to local muscles as a result of pain, previous injury or chronic dysfunction. The broad concept of SMCD suggests that it may be necessary to break or manage a dysfunctional pattern prior to exercises and techniques designed to improve stability or motor control.

It is also necessary to identify the level and involvement of SMCD. Subcategories of SMCD can be organized into two levels: static and dynamic. This demonstrates the hierarchy of motor control, which develops from static postural control to dynamic postural control. To demonstrate functional motor control, it is necessary to establish static and dynamic stabilization; however, it is more important to identify compensatory behaviors that can provide pseudo-stabilization. Inefficient breathing, anxiety breathing, and high threshold strategies are examples of compensation behavior that can appear functional on the surface and with basic strength testing, but would still be considered dysfunctional. Excessive activity and tone in global musculature may also indicate local muscular dysfunction. The presence of neuromuscular trigger points in muscles that do not appear to be high tone or in a shortened state may contribute to poor stabilization as appropriate timing and coordination may be compromised.

The following are just some examples of stability/motor control dysfunctions:

- Motor Control Dysfunction
- Mechanical breathing dysfunction
- High threshold strategy
- Prime mover or global muscle compensation behavior or asymmetry
- Local muscle dysfunction or asymmetry
- Poor static stabilization, alignment, postural control, asymmetry and structural integrity
- Poor dynamic stabilization, alignment, postural control, asymmetry and structural integrity

SFMA TERMINOLOGY



The SFMA breakout flowcharts are designed to separate pain and dysfunction when possible. They will help you identify movement patterns where exercise is indicated or contraindicated.

Unidentified mobility problems must be ruled out as underlying causes of SMCD. This forces us to always consider the regional interdependence model where limited mobility at one moving segment requires a distortion of motor control at nearby segments. Therefore, it is paramount in all SMCD scenarios to identify the compensatory behavior as well as the dysfunction.

Joint stiffness, high muscle tone, trigger points and all other forms of dysfunctional mobility may actually be compensations and natural responses in the presence of poor motor control. This creates the chicken or egg scenario, which can be confusing, but the SFMA provides a logical management tool.

First, reduce or remove compensatory behavior and manage limited mobility. Demonstrate a measurable change in mobility if a limitation is initially present. Then, introduce activities and exercises that reestablish motor control at a level consistent with the SFMA functional status. This means exercises should be used at or below the functional level of the SFMA breakout test. The flowcharts simply suggest that patterns involving pain should be treated, but not exercised, and patterns involving dysfunction without pain should be exercised.

Special circumstances may warrant exercise in some cases, but most models suggest that motor control will be distorted, and outcomes will be inconsistent when exercise is performed in the presence of pain. The suggestions in each flowchart instruct you to look at the subcategories, and then confirm each with proper testing. These tests fall within the scope and responsibility of musculoskeletal evaluation and treatment, but are not part of the SFMA flowchart.

Proper tests can include, but are not limited to:

- Neurological testing for sensory motor integrity
- Muscle strength testing
- Joint stability testing
- Joint mobility testing
- Tissue tension testing – neurological structures, fascial structures, etc.
- Identification of neuromuscular trigger points
- Impairment measurements – goniometry, circumferential measurements, etc.